PES University, Bengaluru

(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - JULY - 2023

UE20CS352 - Object Oriented Analysis & Design with Java

Total Marks : 100.0

1.a. Write briefly, with appropriate examples, about the Analysis and Design phases in Object Oriented Approach for software development. Distinguish this approach from the Procedure oriented approach. (6.0 Marks)

1.b. Identify the use cases and actors and their relationships for Passport Automation System and depict it with a Use Case diagram:

Passport Automation System: To simplify the process of applying for a passport, a software system has to be created. Initially the applicant will login to the passport automation system and submit his details. These details are stored in the database and verification process is done by the passport administrator, the regional administrator and the police, then the passport is issued to the applicant.
- Passport Automation System is used in the effective dispatch of passport to all of the applicants. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in a cogent manner.
- The core of the system is to get the online registration form (with details such as name, address etc.,) filled by the applicant whose testament is verified for its genuineness by the Passport Automation System with respect to the already existing information in the database.
- This forms the first and foremost step in the processing of passport application. After the first round of verification done by the system, the information is in turn forwarded to the regional administrator's (Ministry of External Affairs) office.
- The applicant can optionally seek ECNR status enabled by providing Graduation Certificates and Employment details. This is processed by the Passport Administrator.
- The application is then processed manually based on the report given by the system, and any forfeiting identified can make the applicant liable to penalty as per the law.
- The system forwards the necessary details to the police for its separate verification whose report is then presented to the administrator. After all the necessary criteria have been met, the original information is added to the database and the passport is sent to the applicant. (8.0 Marks)

1.c. What are the different elements in a Deployment Diagram? Write the notations for these in an example diagram. (6.0 Marks)

2.a. Write about the three access specifiers i.e. private, protected and public, that can be used in Java. Using one example class, briefly write about the effect of using the access specifiers on member variables. (6.0 Marks)

2.b. Write code to add the following functionality expected by the MainClass. The Employee object can be created in two ways. The Manager is an Employee. In addition, the Manager has an array/list of Employees who are his/her team members.

```
class Employee{
  private  String name;
  private  String dept;

  //Write code here
}

// Write code here

class  MainClass{
  //Do not change this method
  public static void main(String[] args) {
     Employee e1 = new Employee("Employee Name");
     Employee e2 = new Employee ("Employee Name", " Employee Department");

     Manager m1 = new Manager(“Manager Name”, “Manager Department”);
     m1.addTeamMember(e1);
     m1.addTeamMember(e2);

     m1.showDetails(); // shows details of Manager as well as all his/her team
members
  }
}
```

Note: only write the code expected. Do not repeat the given code. (6.0 Marks)

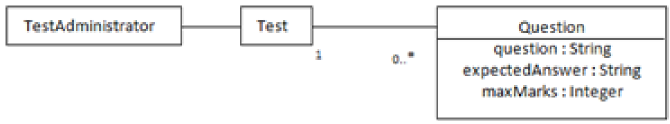
2.c. Write short notes about the following. Use code examples as applicable.

1. Method overloading (2 marks)
2. Static and Instance members in a class (3 marks)
3. Abstract class (3 marks)

(8.0 Marks)

3.a. In the given class model, the responsibility to compute the total marks a Test is being administered for, should be assigned to which class? The assignment of method is based on which principle? Write Java code snippet for the same.

(6.0 Marks)



3.b. State the Interface Segregation Principle. Explain the the problem and solution, with an appropriate example. (8.0 Marks)

3.c. Describe briefly the Pure Fabrication GRASP Design Principle, with an example. (6.0 Marks)

4.a. Write briefly about the Adapter Structural Design Pattern. Clearly state the intent, problem , solution structure and example. (6.0 Marks)

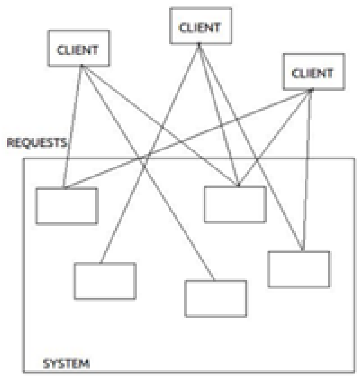
4.b. Implement the following use case using a proxy pattern. Write both the class diagram and the code snippet. State which type of proxy pattern is suitable for the given use case.

Use Case:
A department has thousands of students, with all their details Name, SRN, Dept, etc. At any given time, it is very likely that the user needs the grades of only one student. The solution displays only the *Name*, *SRN*, and *Dept* of the students when a list of them is displayed. When a student is selected, the grades of the corresponding student will be fetched from the database and displayed.

Note: do not write the entire code; write only code snippets/pseudo code to instantiate and use the proxy and the real object. (6.0 Marks)

4.c. What is the problem with the following system? What design pattern can be used to overcome that problem? Write the structure of the solution. State the advantages of using that solution.

(8.0 Marks)



5.a. Explain the Chain of Responsibilities behavioral pattern. State the intent, structure and benefits of this pattern. (6.0 Marks)

5.b. What is the refactored solution to tackle Vendor Lock-in Anti-pattern?

(6.0 Marks)

5.c. Mention symptoms and typical causes for the Blob design anti-pattern.

(8.0 Marks)